

Number Theory and Cryptography(密码学)

Chapter 4

Divisibility and Modular Arithmetic

Section 4.1

Division

Definition: If a and b are integers with $a \neq 0$, then a *divides* b (a 整除 b) if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor or divisor* (因子) of b and that b is a *multiple* (倍数) of a .
- The notation $a \mid b$ denotes that a divides b .
- If a does not divide b , we write $a \nmid b$.

Example: Determine whether $3 \mid 7$ and whether $3 \mid 12$.

$$3 \nmid 7 \quad 3 \mid 12$$

Properties of Divisibility

Theorem 1: Let a , b , and c be integers, where $a \neq 0$.

- i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- ii. If $a \mid b$, then $a \mid bc$ for all integers c ;
- iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Corollary: If a , b , and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

4.1.3 Division Algorithm

When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the “Division Algorithm,” but is really a theorem.

Division Algorithm: If a is an integer and d a positive integer, then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$ (proved in Section 5.2).

- d is called the **divisor**(除数).
- a is called the **dividend**(被除数).
- q is called the **quotient**(商).
- r is called the **remainder**(余数).

Definitions of Functions
div and **mod**

$$q = a \text{ div } d$$

$$r = a \text{ mod } d$$

Examples:

- What are the quotient and remainder when 101 is divided by 11?
- **Solution:** $101 = 11 \cdot 9 + 2$. The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.
- What are the quotient and remainder when -11 is divided by 3?
- **Solution:** $-11 = 3 \cdot (-4) + 1$. The quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$. We don't use this equation $-11 = 3 \cdot (-3) - 2$, since $r = -2$ does not satisfy $0 \leq r < 3$.

4.1.4 Modular Arithmetic(模算数)

Congruence Relation(同余关系)

Definition: If a and b are integers and m is a positive integer, then a is congruent to b modulo m (a 模 m 同余 b) if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a *congruence* (同余式) and that m is its *modulus* (模数).
- If a is not congruent to b modulo m , we write $a \not\equiv b \pmod{m}$.

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution: $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.

- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

The Relationship between (mod m) and **mod** m Notations

The use of “mod” in $a \equiv b \pmod{m}$ and $a \bmod m = b$ are different.

- $a \equiv b \pmod{m}$ is a **relation** on the set of integers.
($1 \equiv 3 \equiv 5 \pmod{2}$)
- In $a \bmod m = b$, the notation **mod** denotes a **function**.
($3 \bmod 2 = 1$)

The relationship between these notations is made clear in this theorem.

Theorem 3: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

More on Congruences

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
 - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
 - $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Computing the **mod** m Function of Products and Sums

Corollary 2 shows how to find the values of the **mod** m function at the sum and product of two integers using the values of this function at each of these integers.

Corollary 2: Let m be a positive integer and let a and b be integers. Then

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

and

$$ab \bmod m = ((a \bmod m) (b \bmod m)) \bmod m.$$

(proof in text)

37. Find each of these values.

a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

27. Evaluate these quantities.

a) $13 \bmod 3$

b) $-97 \bmod 11$

29. Find $a \operatorname{div} m$ and $a \bmod m$ when

a) $a = 228, m = 119$.

35. Decide whether each of these integers is congruent to 5 modulo 17.

a) 80

b) 103

Section Summary₁

1, Division $a \mid b$

2, Division Algorithm $a = dq + r$

- Definitions of Functions **div** and **mod**

$$q = a \text{ div } d$$

$$r = a \text{ mod } d$$

3, Modular Arithmetic

If a and b are integers and m is a positive integer, then a is congruent to b modulo m (a 模 m 同余 b) if m divides $a - b$. $a \equiv b \pmod{m}$.

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Integer Representations and Algorithms

Section 4.2

Representations of Integers

In the modern world, we use *decimal*, or *base 10, notation* to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.

We can represent numbers using any base b , where b is a positive integer greater than 1.

The bases $b = 2$ (*binary*), $b = 8$ (*octal*), and $b = 16$ (*hexadecimal*) are important for computing and communications

Base b Representations

We can use positive integer b greater than 1 as a base, because of this theorem:

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer, $a_0, a_1, \dots, a_k$ are nonnegative integers less than b , and $a_k \neq 0$. The $a_j, j = 0, \dots, k$ are called the base- b digits of the representation.

(We will prove this using mathematical induction in Section 5.1.)

The representation of n given in Theorem 1 is called the *base b expansion of n* and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.

We usually omit the subscript 10 for base 10 expansions.

Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

Octal Expansions

The octal expansion (base 8) uses the digits $\{0,1,2,3,4,5,6,7\}$.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Hexadecimal Expansions

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}. The letters A through F represent the decimal numbers 10 through 15.

Example: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution:

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

Example: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution: $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$

Base Conversion₁

$$895 = 10 \cdot 89 + 5$$

$$89 = 10 \cdot 8 + 9$$

$$8 = 10 \cdot 0 + 8$$

To construct the base b expansion of an integer n :

- Divide n by b to obtain a quotient and remainder.

$$n = bq_0 + a_0 \quad 0 \leq a_0 \leq b$$

- The remainder, a_0 , is the rightmost digit in the base b expansion of n . Next, divide q_0 by b .

$$q_0 = bq_1 + a_1 \quad 0 \leq a_1 \leq b$$

- The remainder, a_1 , is the second digit from the right in the base b expansion of n .
- Continue by successively dividing the quotients by b , obtaining the additional base b digits as the remainder. The process terminates when the quotient is 0.

Construction Base b Expansion (optional)

$$895 = 10 \cdot 89 + 5$$

$$89 = 10 \cdot 8 + 9$$

$$8 = 10 \cdot 0 + 8$$

procedure *base b expansion*(n, b : positive integers with $b > 1$)

$q := n$

$k := 0$

while ($q \neq 0$)

$a_k := q \bmod b$

$q := q \text{ div } b$

$k := k + 1$

return($a_{k-1}, \dots, a_1, a_0$) $\{(a_{k-1} \dots a_1 a_0)_b$ is base b expansion of $n\}$

q represents the quotient obtained by successive divisions by b , starting with $q = n$.

The digits in the base b expansion are the remainders of the division given by $q \bmod b$.

The algorithm terminates when $q = 0$ is reached.

Base Conversion₂

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

- $12345 = 8 \cdot 1543 + 1$
- $1543 = 8 \cdot 192 + 7$
- $192 = 8 \cdot 24 + 0$
- $24 = 8 \cdot 3 + 0$
- $3 = 8 \cdot 0 + 3$

The remainders are the digits from right to left yielding $(30071)_8$.

Comparison of Hexadecimal, Octal, and Binary Representations

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Initial 0s are not shown

Each octal digit corresponds to a block of 3 binary digits.

Each hexadecimal digit corresponds to a block of 4 binary digits.

So, conversion between binary, octal, and hexadecimal is easy.

Conversion Between Binary, Octal, and Hexadecimal Expansions

Example: Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.

Solution:

- To convert to octal, we group the digits into blocks of three $(011\ 111\ 010\ 111\ 100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, 7, 2, 7, and 4. Hence, the solution is $(37274)_8$.
- To convert to hexadecimal, we group the digits into blocks of four $(0011\ 1110\ 1011\ 1100)_2$, adding initial 0s as needed. The blocks from left to right correspond to the digits 3, E, B, and C. Hence, the solution is $(3EBC)_{16}$.

- **Binary addition**

Add two binary numbers 10011011 and 1011011.

$$\begin{array}{r} 10011011 \\ + 1011011 \\ \hline 11110110 \end{array}$$

- **Hexadecimal Addition**

Add the two hexadecimal numbers 84F and 42EA

$$84F + 42EA = ?$$

$$F + A = 9 \text{ carry } 1$$

$$\text{Now } 4 + E + 1(\text{carry}) = 3 \text{ carry } 1$$

$$\text{Next } 8 + 2 + 1(\text{carry}) = 11 \text{ carry } 0$$

$$\text{Finally } 4 + 0(\text{carry}) = 4$$

Hence addition result is $4B39_{16}$.

Section Summary₂

Integer Representations

- Base b Expansions
- Binary Expansions
- Octal Expansions
- Hexadecimal Expansions

Base Conversion Algorithm

1. Convert the decimal expansion of each of these integers to a binary expansion.

a) 231

3. Convert the binary expansion of each of these integers to a decimal expansion.

a) $(1\ 1111)_2$

5. Convert the octal expansion of each of these integers to a binary expansion.

a) $(572)_8$

Convert Decimal to Hexadecimal

$$10958_{10} = 2ACE_{16}?$$

21. Find the sum

a) $(100\ 0111)_2$, $(111\ 0111)_2$

Primes and Greatest Common Divisors

Section 4.3

4.3.2 Primes(素数)

Definition: A positive integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*(合数).

Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

The Fundamental Theorem of Arithmetic

Theorem: Every positive integer greater than 1 can be written **uniquely** as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

Examples:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$
- $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

4.3.6 Greatest Common Divisors and Least Common Multiples

Greatest Common Divisor₁

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the **greatest common divisor**(最大公约数) of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

One can find greatest common divisors of small numbers by inspection.

Example: What is the greatest common divisor of 24 and 36?

Solution: The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $\gcd(24, 36) = 12$.

Example: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17, 22) = 1$

Greatest Common Divisor₂

Definition: The integers a and b are *relatively prime* (互素) if their greatest common divisor is 1.

Example: 17 and 22 are relatively prime.

Definition: The integers $a_1, a_2, \dots, a_n$ are *pairwise relatively prime* (两两互素) if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Finding the Greatest Common Divisor Using Prime Factorizations(素数分解)

Suppose the prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n},$$

where each exponent(指数) is a nonnegative integer, and where all primes occurring in either prime factorization are included in both.

Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)},$$

This formula is valid since the integer on the right (of the equals sign) divides both a and b . No larger integer can divide both a and b .

Example: $120 = 2^3 \cdot 3 \cdot 5$ $500 = 2^2 \cdot 5^3$

$$\gcd(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$$

Finding the gcd of two positive integers using their prime factorizations is not efficient because there is no efficient algorithm for finding the prime factorization of a positive integer.

Least Common Multiple

Definition: The **least common multiple** (最小公倍数) of the positive integers a and b is the smallest positive integer that is divisible by both a and b . It is denoted by $\text{lcm}(a,b)$.

The least common multiple can also be computed from the prime factorizations.

$$\text{lcm}(a,b) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)},$$

This number is divided by both a and b and no smaller number is divided by a and b .

Example: $\text{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3,4)} 3^{\max(5,3)} 7^{\max(2,0)} = 2^4 3^5 7^2$

The greatest common divisor and the least common multiple of two integers are related by:

Theorem 5: Let a and b be positive integers. Then

$$ab = \text{gcd}(a,b) \cdot \text{lcm}(a,b)$$

4.3.7 Euclidean Algorithm₁

The **Euclidian algorithm** is an efficient method for computing the **greatest common divisor** of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(b,c)$ when $a > b$ and c is the remainder when a is divided by b .



Euclid
(325 B.C.E. – 265 B.C.E.)

Example: Find $\gcd(91, 287)$:

- $287 = 91 \cdot 3 + 14$ Divide 287 by 91
- $91 = 14 \cdot 6 + 7$ Divide 91 by 14
- $14 = 7 \cdot 2 + 0$ Divide 14 by 7
Stopping condition

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

Correctness of Euclidean Algorithm₁

Lemma 1: Let $a = bq + r$, where a , b , q , and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

Proof:

- Suppose that d divides both a and b . Then d also divides $a - bq = r$ (by Theorem 1 of Section 4.1). Hence, any common divisor (公因子) of a and b must also be any common divisor of b and r .
- Suppose that d divides both b and r . Then d also divides $bq + r = a$. Hence, any common divisor of b and r must also be a common divisor of a and b .
- Therefore, $\gcd(a, b) = \gcd(b, r)$.

Correctness of Euclidean Algorithm₂

Suppose that a and b are positive integers with $a \geq b$.

Let $r_0 = a$ and $r_1 = b$.

Successive applications of the division algorithm yields:

$$r_0 = r_1 q_1 + r_2 \quad 0 \leq r_2 < r_1,$$

$$r_1 = r_2 q_2 + r_3 \quad 0 \leq r_3 < r_2,$$

.

.

.

$$r_{n-2} = r_{n-1} q_{n-1} + r_n \quad 0 \leq r_n < r_{n-1},$$

$$r_{n-1} = r_n q_n.$$

Eventually, a remainder of zero occurs in the sequence of terms: $a = r_0 > r_1 > r_2 > \cdots \geq 0$. The sequence can't contain more than a terms.

By Lemma 1

$$\gcd(a, b) = \gcd(r_0, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n.$$

Hence the **greatest common divisor is the last nonzero remainder** in the sequence of divisions.

Euclidean Algorithm₂

The Euclidean algorithm expressed in pseudocode is:

```
procedure gcd(a, b: positive integers)
```

```
  x := a                                287
```

```
  y := b                                91
```

```
  while y ≠ 0
```

```
    r := x mod y                        14      7      0
```

```
    x := y                              91      14      7
```

```
    y := r                              14      7      0
```

```
  return x {gcd(a,b) is x}
```

Example: Find gcd(91, 287):

- $287 = 91 \cdot 3 + 14$

- $91 = 14 \cdot 6 + 7$

- $14 = 7 \cdot 2 + 0$

Example: Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

Solution: Successive uses of the division algorithm give:

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41.$$

Hence, $\gcd(414, 662) = 2$, because 2 is the last nonzero remainder.

We can summarize these steps in tabular form.

j	r_j	r_{j+1}	q_{j+1}	r_{j+2}
0	662	414	1	248
1	414	248	1	166
2	248	166	1	82
3	166	82	2	2
4	82	2	41	0

4. Find the prime factorization of each of these integers.

a) 39

b) 81

16. Determine whether the integers in each of these sets are pairwise relatively prime.

a) 21, 34, 55

b) 14, 17, 85

25. What are the greatest common divisors of these pairs of integers?

a) $3^7 \cdot 5^3 \cdot 7^3$, $2^{11} \cdot 3^5 \cdot 5^9$

27. What is the least common multiple of each pair in Exercise 25?

- 33.** Use the Euclidean algorithm to find
- a)** $\gcd(12, 18)$.

Section Summary₃

1, Prime Numbers and their Properties

2, Greatest Common Divisors and Least Common Multiples

$\gcd(a,b)$ $\text{lcm}(a,b)$

3, The Euclidian Algorithm

Lemma 1: Let $a = bq + r$, where a , b , q , and r are integers. Then $\gcd(a,b) = \gcd(b,r)$.

Chapter Summary

Divisibility and Modular Arithmetic

Primes and Greatest Common Divisors

Homework

Page 259 26ab, 28a, 34ac

Page 269 6a. 7a. 23a.(just calculate the sum)

Page 288 32c.

1. Give a big- O estimate for the number of operations (where an operation is an addition or a multiplication) used in this segment of an algorithm.

$t := 0$

for $i := 1$ **to** 3

for $j := 1$ **to** 4

$t := t + ij$

$O(1)$

$i=1$ 4次加法 $i=2$ 4次加法 $i=3$ 四次加法

$t = 0 + 1*1 + 1*2 + 1*3 + 1*4 + 2*1 + 2*2 + 2*3 + 2*4 + 3*1 + 3*2 + 3*3 + 3*4$

4. Give a big- O estimate for the number of operations, where an operation is an addition or a multiplication, used in this segment of an algorithm (ignoring comparisons used to test the conditions in the **while** loop).

$i := 1$

$t := 0$

while $i \leq n$

$t := t + i$

$i := 2i$

$O(\log n)$

$i=1$ $i=2$ $i=4$ $i=8$ $i=16$ $i=32$ $i=2^{\{k\}} < n$ $2^{\{k\}} < n$ $k = \log_2 n$

$t = 1 + 2 + 4 + 8 + 16 + 32 + \dots + 2^{\{k\}}$

$k = \lfloor \log_2 n \rfloor$

15. What is the largest n for which one can solve within one second a problem using an algorithm that requires $f(n)$ bit operations, where each bit operation is carried out in 10^{-9} seconds, with these functions $f(n)$?

- | | | |
|-------------|----------|---------------|
| a) $\log n$ | b) n | c) $n \log n$ |
| d) n^2 | e) 2^n | f) $n!$ |

- a) 一秒 可以运算 10^9 次 $n=?$ $\log n = 10^9$ $n = 2^{10^9}$
 b) $n=?$ $n = 10^9$ $n = 10^9$
 c) D) $n=?$ $n^2 = 10^9$ $n = 10^{\{4.5\}}$ $n = 31622$

3. Give a big- O estimate for the number of operations, where an operation is a comparison or a multiplication, used in this segment of an algorithm (ignoring comparisons used to test the conditions in the **for** loops, where $a_1, a_2, \dots, a_n$ are positive real numbers).

```

m := 0
for i := 1 to n
    for j := i + 1 to n
        m := max( $a_i a_j$ , m)
    
```

算乘法的数目 或者算比较的次数 循环一次 有一次运算

$$n-1 + n-2 + n-3 + \dots + 2 + 1 = (n-1+1)/2 * (n-1) = n^2 - n \quad O(n^2)$$